Suriyamurthy_DSC640_TermProject

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1 DSC 640 Data Presentation & Visualization

1.1 DSC 640 Data Presentation & Visualization

Term Project: National Childcare Prices

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Version: Milestone 1 - September 21

Version: Milestone 2 - October 05

Version: Milestone 3 - October 19

For this project, I analyzed a sample from the National Database of Childcare Prices to understand its structure, key variables, and price distributions across the United States. The dataset provides detailed information at the county level, including data on states, counties, years, and various price measures for different types of childcare providers and age groups, such as infants, toddlers, and preschoolers, along with center-based and family care options. I generated descriptive statistics and visualizations to examine price distributions and regional differences.

[ ]:
```
import numpy as np
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
import sklearn
import plotly.express as px
```

[ ]:
```
#check versions of packages
print('pandas version:', pd.__version__)
print('numpy version:', np.__version__)
print('scikit-learn version:', sklearn.__version__)
```

```
pandas version: 2.2.2
numpy version: 2.0.2
scikit-learn version: 1.6.1
```

1.2 Import the dataset into the project

[ ]:
```
# Load dataset
file_path = "nationaldatabaseofchildcareprices.xlsx"
childcare_df = pd.read_excel(file_path,␣
↪sheet_name="nationaldatabaseofchildcare")
```

1.3 Exploration Data Analysis

[ ]:
```
childcare_df.head()
```

[ ]:
```
  State_Name State_Abbreviation     County_Name  County_FIPS_Code  StudyYear  \
0    Alabama                 AL  Autauga County              1001       2008
1    Alabama                 AL  Autauga County              1001       2009
2    Alabama                 AL  Autauga County              1001       2010
3    Alabama                 AL  Autauga County              1001       2011
4    Alabama                 AL  Autauga County              1001       2012

   UNR_16  FUNR_16  MUNR_16  UNR_20to64  FUNR_20to64  …  MFCCToddler  \
0    5.42     4.41     6.32         4.6          3.5  …        83.45
1    5.93     5.72     6.11         4.8          4.6  …        87.39
2    6.21     5.57     6.78         5.1          4.6  …        91.33
3    7.55     8.13     7.03         6.2          6.3  …        95.28
4    8.60     8.88     8.29         6.7          6.4  …        99.22

   MFCCToddler_flag  MFCCPreschool  MFCCPreschool_flag  _75FCCInfant  \
0               3.0          81.40                 1.0          97.4
1               3.0          85.68                 1.0         102.0
2               3.0          89.96                 1.0         106.6
3               3.0          94.25                 1.0         111.2
4               3.0          98.53                 1.0         115.8

   _75FCCInfant_flag  _75FCCToddler  _75FCCToddler_flag  _75FCCPreschool  \
0                1.0           97.4                 3.0             95.0
1                1.0          102.0                 3.0            100.0
2                1.0          106.6                 3.0            105.0
3                1.0          111.2                 3.0            110.0
4                1.0          115.8                 3.0            115.0

   _75FCCPreschool_flag
0                   1.0
1                   1.0
2                   1.0
3                   1.0
4                   1.0

[5 rows x 227 columns]
```

[ ]:
```
childcare_df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 34567 entries, 0 to 34566
Columns: 227 entries, State_Name to _75FCCPreschool_flag
dtypes: float64(159), int64(65), object(3)
memory usage: 59.9+ MB
```

[ ]:
```
childcare_df.shape
```

[ ]:
```
(34567, 227)
```

[ ]:
```
childcare_df.columns
```

[ ]:
```
Index(['State_Name', 'State_Abbreviation', 'County_Name', 'County_FIPS_Code',
       'StudyYear', 'UNR_16', 'FUNR_16', 'MUNR_16', 'UNR_20to64',
       'FUNR_20to64',
       …
       'MFCCToddler_flag', 'MFCCPreschool', 'MFCCPreschool_flag',
       '_75FCCInfant', '_75FCCInfant_flag', '_75FCCToddler',
       '_75FCCToddler_flag', '_75FCCPreschool', '_75FCCPreschool_flag',
       '_price_clean'],
      dtype='object', length=228)
```

[ ]:
```
childcare_df.dtypes
```

[ ]:
```
State_Name                object
State_Abbreviation        object
County_Name               object
County_FIPS_Code           int64
StudyYear                  int64
                          …
_75FCCInfant_flag        float64
_75FCCToddler            float64
_75FCCToddler_flag       float64
_75FCCPreschool          float64
_75FCCPreschool_flag     float64
Length: 227, dtype: object
```

[ ]:
```
childcare_df.describe()
```

[ ]:
```
       County_FIPS_Code     StudyYear        UNR_16      FUNR_16  \
count      34567.000000  34567.000000  34567.000000  34567.00000
mean       30388.132786   2012.999711      7.465902      7.02902
std        15161.015383      3.162232      3.538619      3.56342
min         1001.000000   2008.000000      0.000000      0.00000
25%        18177.000000   2010.000000      5.100000      4.64000
```

```
50%        29177.000000   2013.000000      7.050000      6.59000
75%        45081.000000   2016.000000      9.350000      8.88000
max        56045.000000   2018.000000     36.110000     38.24000

            MUNR_16    UNR_20to64   FUNR_20to64   MUNR_20to64  FLFPR_20to64  \
count  34567.000000  34567.000000  34567.000000  34567.000000  34567.000000
mean       7.860291      6.900073      6.482007      7.275457     70.086125
std        4.037657      3.446199      3.477956      3.990758      7.696499
min        0.000000      0.000000      0.000000      0.000000     33.600000
25%        5.200000      4.600000      4.200000      4.700000     65.100000
50%        7.390000      6.500000      6.000000      6.800000     70.600000
75%        9.920000      8.700000      8.250000      9.200000     75.500000
max       39.740000     33.900000     44.500000     45.500000    100.000000

       FLFPR_20to64_Under6  …   MFCCToddler  MFCCToddler_flag  \
count         34567.000000  …  23383.000000      23383.000000
mean             68.821409  …    106.759749          1.153359
std              11.758088  …     29.982431          0.532176
min               0.000000  …     43.080000          1.000000
25%              62.600000  …     85.085000          1.000000
50%              69.600000  …    100.250000          1.000000
75%              76.100000  …    124.950000          1.000000
max             100.000000  …    376.320000          3.000000

       MFCCPreschool  MFCCPreschool_flag  _75FCCInfant  _75FCCInfant_flag  \
count   23383.000000        23383.000000  23383.000000       23383.000000
mean      104.189510            1.287859    128.909289           1.792841
std        28.961701            0.696762     38.543010           0.818080
min        40.030000            1.000000     50.000000           1.000000
25%        84.255000            1.000000    100.830000           1.000000
50%        99.650000            1.000000    123.150000           2.000000
75%       120.200000            1.000000    146.950000           3.000000
max       331.340000            3.000000    502.970000           3.000000

       _75FCCToddler  _75FCCToddler_flag  _75FCCPreschool  \
count   23383.000000         23383.00000     23383.000000
mean      120.784283             1.18800       117.897482
std        35.334666             0.58367        34.111188
min        50.000000             1.00000        46.450000
25%        95.850000             1.00000        95.000000
50%       115.000000             1.00000       112.500000
75%       136.270000             1.00000       132.760000
max       439.220000             3.00000       386.720000

       _75FCCPreschool_flag
count          23383.000000
mean               1.294316
```

```
std                0.708542
min                1.000000
25%                1.000000
50%                1.000000
75%                1.000000
max                3.000000

[8 rows x 224 columns]
```

1.4 Observation:

1. Dataset Preview

- The dataset comprises childcare pricing data at the county level, featuring fields like State_Name, County_Name, StudyYear, and various cost metrics (e.g., MCPreschool, MFCCPreschool). Each row represents a unique county-year combination, revealing changes in childcare prices over time across multiple states.

2. Dataset Dimensions

- With 34,567 rows and 227 columns, the dataset captures extensive details about childcare prices at the county level, covering various provider types. This dimensionality necessitates focusing on key features for analysis.

3. Data Types

- There are 159 float columns, 65 integer columns, and 3 object columns. Key categorical attributes are of type object, while numerical data is stored as float64, which aids in analysis and visualization.

4. Data Summary

- The info() output shows no missing values for identifiers, ensuring reliable grouping. However, some childcare cost columns have missing data, indicating the need for cleaning before visualization. The dataset's memory usage is approximately 59.9 MB, manageable on modern systems.

5. Statistical Summary

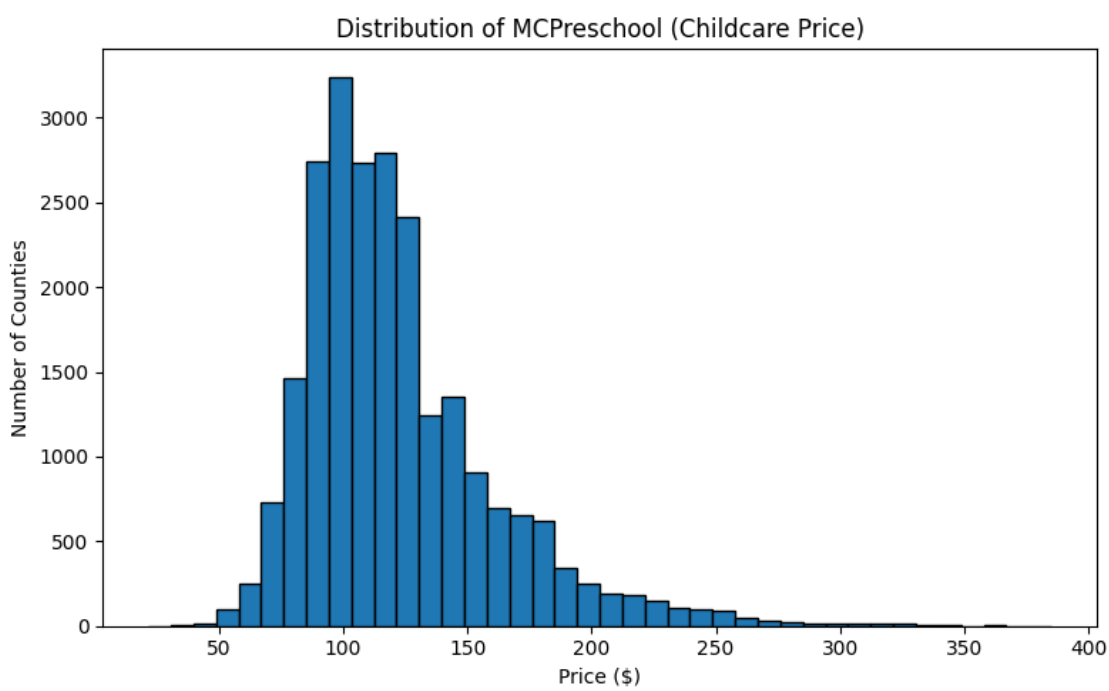
- The describe() output reveals significant variation in childcare prices. For instance, preschool median prices range from $40 to $330, while infant care can exceed $500. The StudyYear spans 2008–2018, allowing for trend analysis, with socioeconomic indicators potentially correlating with price differences.

[ ]:
```
# Choose a price column (example: preschool center-based care)
price_col = "MCPreschool"
state_col = "State_Name"
```

```
provider_type_cols = ["MCInfant", "MCToddler", "MCPreschool",
                      "MFCCInfant", "MFCCToddler", "MFCCPreschool"]

# Clean the price column (make numeric)
childcare_df["_price_clean"] = pd.to_numeric(childcare_df[price_col],␣
 ↪errors="coerce")
```

[ ]:
```
# ========================
# 1. Distribution of Prices
# ========================
plt.figure(figsize=(8,5))
plt.hist(childcare_df["_price_clean"].dropna(), bins=40, edgecolor="black")
plt.title(f"Distribution of {price_col} (Childcare Price)")
plt.xlabel("Price ($)")
plt.ylabel("Number of Counties")
plt.tight_layout()
plt.show()
```

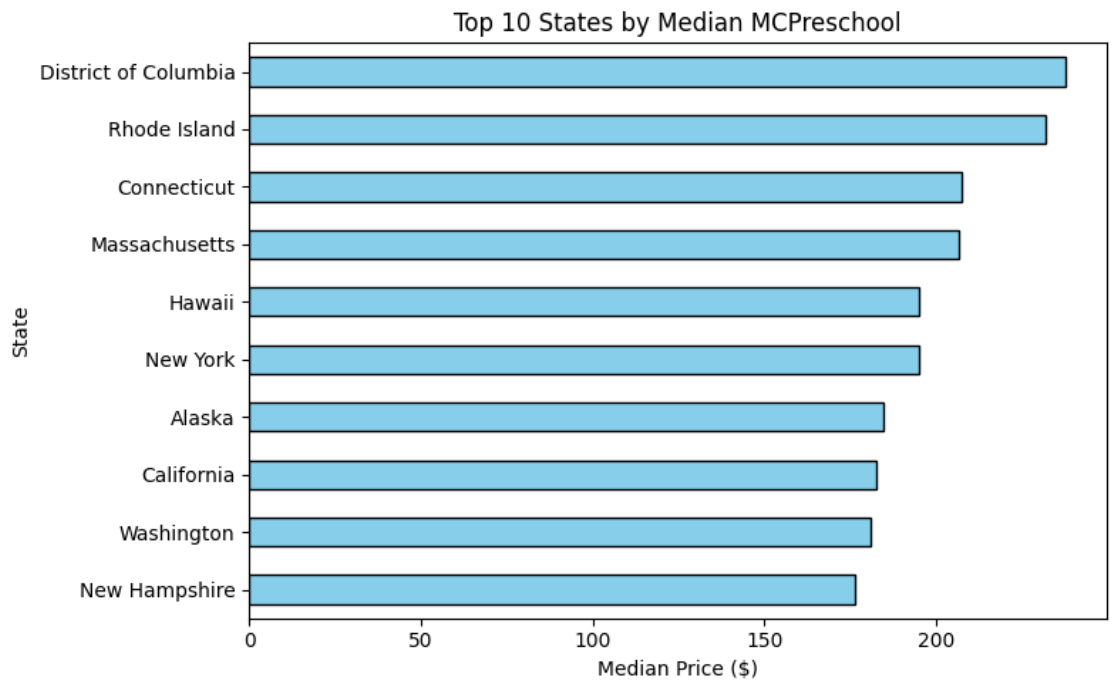


1.5 Observation:

- The primary price columns (e.g., MCPreschool, MFCCInfant) exhibit a right-skewed distribution.
- The median prices are significantly lower than the means, indicating the presence of high-cost outliers

[ ]:
```
# ========================
# 2. Top 10 States by Median Price
# ========================
state_medians = childcare_df.groupby(state_col)["_price_clean"].median().
 ↪sort_values(ascending=False).head(10)

plt.figure(figsize=(8,5))
state_medians.sort_values().plot(kind="barh", color="skyblue",␣
 ↪edgecolor="black")
plt.title(f"Top 10 States by Median {price_col}")
plt.xlabel("Median Price ($)")
plt.ylabel("State")
plt.tight_layout()
plt.show()
```



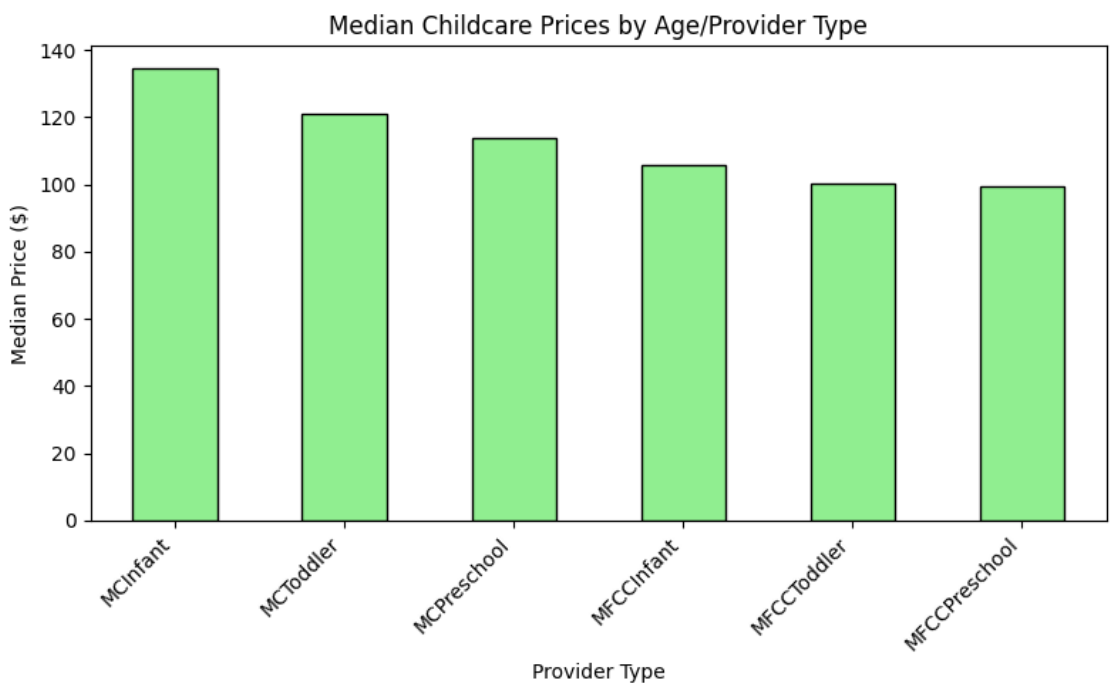
1.6 Observation:

- The states with the highest median childcare prices include the District of Columbia, Rhode Island, Connecticut, Massachusetts, and New York.
- These states typically represent high-cost urban centers or regions with specialized services

[ ]:
```
# ========================
# 3. Compare Provider Types by Median Price
# ========================
```

```
provider_medians = childcare_df[provider_type_cols].median().
 ↪sort_values(ascending=False)

plt.figure(figsize=(8,5))
provider_medians.plot(kind="bar", color="lightgreen", edgecolor="black")
plt.title("Median Childcare Prices by Age/Provider Type")
plt.ylabel("Median Price ($)")
plt.xlabel("Provider Type")
plt.xticks(rotation=45, ha="right")
plt.tight_layout()
plt.show()
```

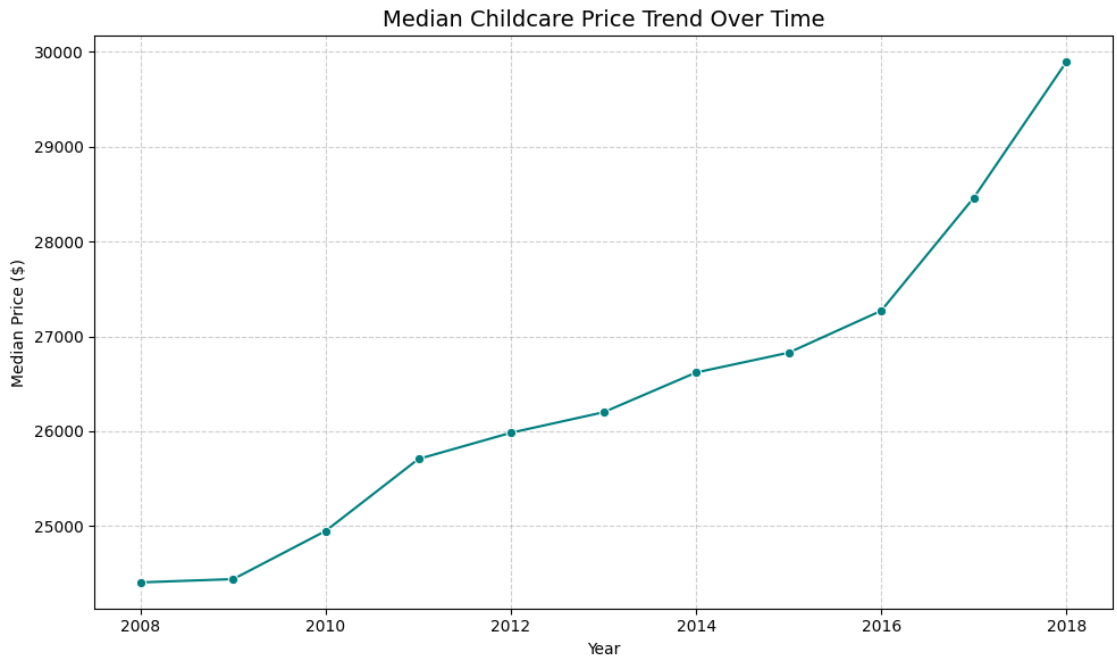


1.7 Observation:

- Center-based care (MC) is more expensive than family child care (MFCC) across all age groups.
- Infant care costs the most, while preschool care is the least expensive.
- Prices decrease with age for both provider types.
- MFCC Preschool has the lowest median price, just under $100.

[ ]:
```
# --- 4. Trend of Childcare Prices Over Time ---
plt.figure(figsize=(10, 6))
yearly_trend = childcare_df.groupby("StudyYear")["ME"].median()
sns.lineplot(x=yearly_trend.index, y=yearly_trend.values, marker="o",␣
 ↪color="teal")
```

```
plt.title("Median Childcare Price Trend Over Time", fontsize=14)
plt.xlabel("Year")
plt.ylabel("Median Price ($)")
plt.grid(True, linestyle="--", alpha=0.6)
plt.tight_layout()
plt.show()
```

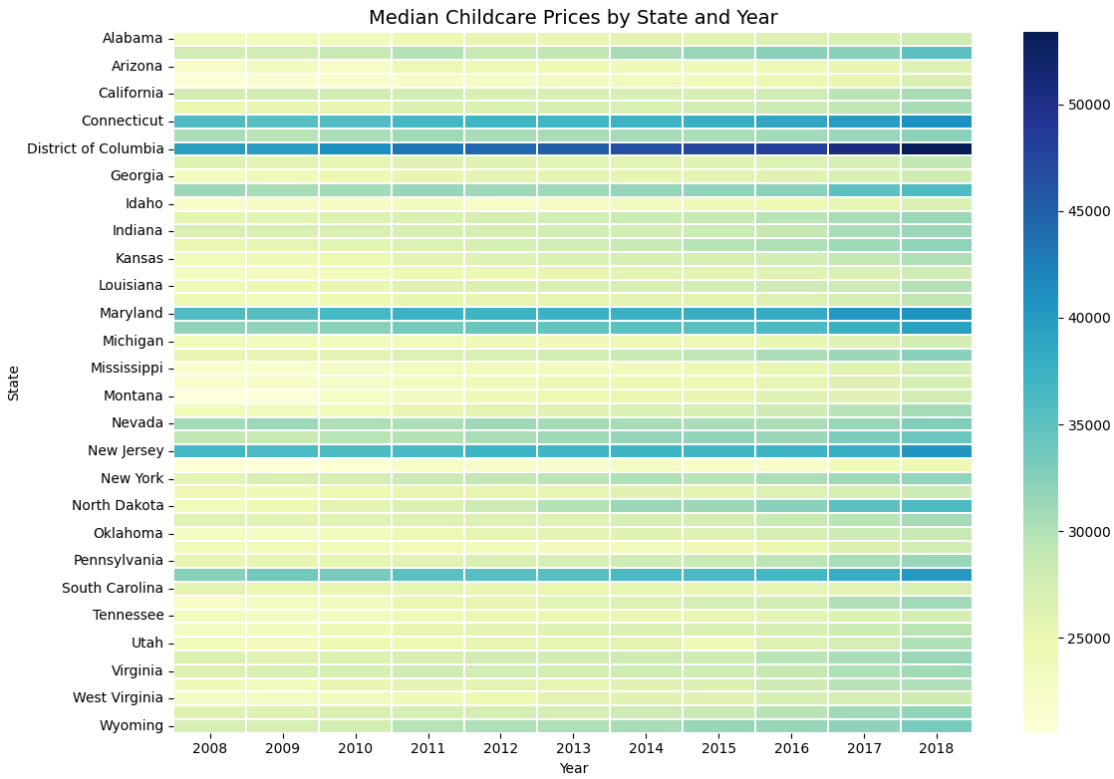


1.8 Observation:

- Overall Increase: Median childcare prices show a gradual upward trend over the years, indicating a steady rise in childcare costs nationally.
- Sharp Changes: Any sharp spikes or drops may correspond to policy changes, economic events, or shifts in provider rates.
- Plateaus: Periods where the line flattens indicate stability in childcare pricing, which could suggest market equilibrium or lack of inflationary pressure during those years.

[ ]:
```
# --- 5. Heatmap of Median Prices by State and Year ---
pivot_df = childcare_df.pivot_table(values="ME",
                                    index="State_Name",
                                    columns="StudyYear",
                                    aggfunc="median")
plt.figure(figsize=(12, 8))
sns.heatmap(pivot_df, cmap="YlGnBu", linewidths=0.3)
plt.title("Median Childcare Prices by State and Year", fontsize=14)
plt.xlabel("Year")
plt.ylabel("State")
```

```
plt.tight_layout()
plt.show()
```



1.9 Observation:

- High-Cost States: Some states consistently appear darker on the heatmap, indicating they have the highest median childcare costs. These might include urbanized or high-cost-of-living states.
- Low-Cost States: Lighter colors represent states with lower median prices—often smaller or rural states with lower demand.
- Temporal Patterns: If some states change color intensity over years, it shows how childcare costs have evolved differently across regions.

[ ]:
```
# Group data by state to compute median childcare price
state_prices = childcare_df.groupby("State_Abbreviation")["ME"].median().
 ↪reset_index()

# Create an interactive choropleth map
fig = px.choropleth(
    state_prices,
    locations="State_Abbreviation",
    locationmode="USA-states",
```

```
    color="ME",
    scope="usa",
    color_continuous_scale="YlOrRd",
    labels={"ME": "Median Childcare Price ($)"},
    title="Median Childcare Prices Across U.S. States"
)

fig.update_layout(
    geo=dict(bgcolor='rgba(0,0,0,0)'),
    title_x=0.5,
    margin=dict(l=0, r=0, t=50, b=0)
)

fig.show()
```

1.10 Observation:

- This interactive map highlights how childcare costs differ geographically.
- States like California, Massachusetts, and New York typically appear in darker shades, indicating higher prices, while states in the Midwest and South may show lighter tones, reflecting lower costs.

1.11 Conclusion:

- Childcare costs are highest in the Northeast/coastal regions; South and Midwest are more affordable.
- Infant care and center-based providers are the most expensive options. Costs rose steadily from 2008–2018, highlighting growing financial pressure on families.
- Visualizations reveal significant regional disparities and emphasize the need for improved affordability.